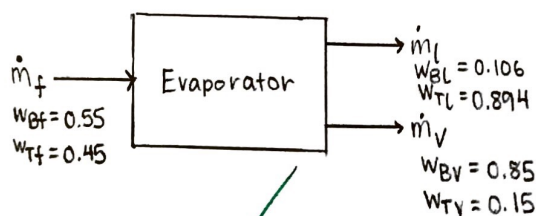


1.

a)

20/20



b) $\dot{m}_f = 100 \text{ kg/hr}$ [Assumption]

Input - Output + Generation - Consumption = Accumulation [st.st.]

$$\dot{m}_f (W_{Bf}) = \dot{m}_L (W_{BL}) + \dot{m}_V (W_{BV})$$

$$100 \frac{\text{kg}}{\text{hr}} (0.55) = \dot{m}_l (0.106) + \dot{m}_v (0.85)$$

$$\text{Input} - \text{Output} + \text{Generation} - \text{Consumption} = \text{Accumulation [st.st.]}$$

$$\dot{m}_f (w_{Tf}) = \dot{m}_l (w_{Tl}) + \dot{m}_v (w_{Tv})$$

$$100 \frac{\text{kg}}{\text{hr}} (0.45) = \dot{m}_l (0.894) + \dot{m}_v (0.15)$$

$$\dot{m}_f = \dot{m}_l + \dot{m}_v$$

$$100 \frac{\text{kg}}{\text{hr}} = \dot{m}_L + \dot{m}_V$$

$$\dot{m}_L = (100 - \dot{m}_V) \frac{\text{kg}}{\text{hr}}$$

$$100 \frac{\text{kg}}{\text{hr}} (0.55) = (100 - \dot{m}_v) \frac{\text{kg}}{\text{hr}} (0.106) + \dot{m}_v (0.85)$$

$$55 \frac{\text{kg}}{\text{hr}} = 10.6 \frac{\text{kg}}{\text{hr}} - 0.106 \dot{m}_v \frac{\text{kg}}{\text{hr}} + 0.85 \dot{m}_v \frac{\text{kg}}{\text{hr}}$$

$$44.4 \frac{\text{kg}}{\text{hr}} = 0.744 \dot{m}_v \frac{\text{kg}}{\text{hr}}$$

$$\dot{m}_v = 59.68 \frac{\text{kg}}{\text{hr}}$$

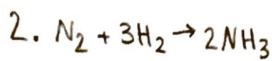
$$\dot{m}_i = (100 - 59.68) \frac{\text{kg}}{\text{hr}}$$

$$\dot{m}_1 = 40.32 \frac{\text{kg}}{\text{hr}}$$

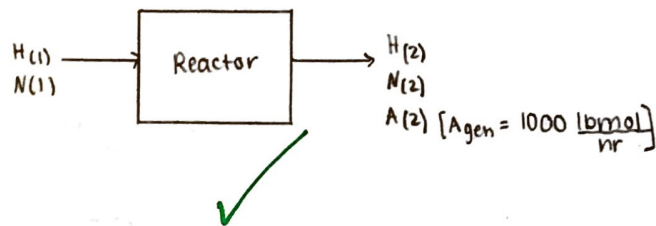
$$\frac{\dot{m}_v}{\dot{m}_l} = \frac{59.68}{40.32} = 1.480$$

c) Stream var: 6 specified flow: 0
System var: $\frac{0}{6}$ specified stream comp.: 3
specified system perf.: 0
material balance eqns.: $\frac{2}{5}$

$$\text{DOF} = 6 - 5 = 1 \text{ [underspecified]}$$



20/20



$T = 1440^\circ R$
 $P = 100 \text{ atm}$
 $25\% N_2 \rightarrow NH_3$

$$\frac{N_{(1)}}{H_{(1)}} = \frac{1 \text{ lbmol } N_2}{3 \text{ lbmol } H_2} = \frac{1}{3}$$

$$\frac{N_{\text{cons}}}{H_{\text{cons}}} = \frac{1 \text{ lbmol } N_2}{3 \text{ lbmol } H_2} = \frac{1}{3}$$

$$\frac{A_{\text{gen}}}{N_{\text{cons}}} = \frac{2 \text{ lbmol } A_2}{1 \text{ lbmol } N_2} = 2$$

$$N_{\text{cons}} = 0.25 N_{(1)}$$

Material Balance for Nitrogen

$$\text{Input} - \text{Output} + \text{Generation} - \text{Consumption} = \text{Accumulation}$$

$$\text{Input} - \text{Output} - \text{Consumption} = 0$$

$$N_{(1)} - N_{\text{cons}} = N_{(2)}$$

Material Balance for Hydrogen

$$\text{Input} - \text{Output} + \text{Generation} - \text{Consumption} = \text{Accumulation}$$

$$\text{Input} - \text{Output} - \text{consumption} = 0$$

$$H_{(1)} - H_{\text{cons}} = H_{(2)}$$

Material Balance for Ammonia

$$\text{Input} - \text{Output} + \text{Generation} - \text{Consumption} = \text{Accumulation}$$

$$-\text{Output} + \text{Generation} = 0$$

$$A_{(2)} = A_{\text{gen}}$$

Total Mass Balance

$$\dot{m}_{(1)} = \dot{m}_{(2)}$$

Solve

$$1000 \frac{\text{lbmol}}{\text{hr}} A_{(2)} \times \frac{1 \text{ lbmol } N_2}{2 \text{ lbmol } A_2} = 500 \frac{\text{lbmol}}{\text{hr}} \rightarrow N_{\text{cons}} = 500 \frac{\text{lbmol}}{\text{hr}}$$

$$N_{\text{cons}} = 0.25 N_{(1)}$$

$$500 \frac{\text{lbmol}}{\text{hr}} = 0.25 N_{(1)}$$

$$N_{(1)} = 2000 \frac{\text{lbmol}}{\text{hr}}$$

$$N_{(1)} - N_{\text{cons}} = N_{(2)}$$

$$2000 \frac{\text{lbmol}}{\text{hr}} - 500 \frac{\text{lbmol}}{\text{hr}} = N_{(2)}$$

$$N_{(2)} = 1500 \frac{\text{lbmol}}{\text{hr}}$$

$$500 \frac{\text{lbmol}}{\text{hr}} N_{\text{cons}} \times \frac{3 \text{ lbmol } H_2}{1 \text{ lbmol } N_2} = 1500 \frac{\text{lbmol}}{\text{hr}}$$

$$H_{\text{cons}} = 1500 \frac{\text{lbmol}}{\text{hr}}$$

$$2000 \frac{\text{lbmol}}{\text{hr}} N_{(1)} \times \frac{3 \text{ lbmol } H_2}{1 \text{ lbmol } N_2} = 6000 \frac{\text{lbmol}}{\text{hr}} \rightarrow H_{(1)} = 6000 \frac{\text{lbmol}}{\text{hr}}$$

$$H_{(1)} - H_{\text{cons}} = H_{(2)}$$

$$6000 \frac{\text{lbmol}}{\text{hr}} - 1500 \frac{\text{lbmol}}{\text{hr}} = H_{(2)}$$

$$H_{(2)} = 4500 \frac{\text{lbmol}}{\text{hr}}$$

$$P\dot{V} = \dot{n}RT \quad \dot{V} = \frac{\dot{n}RT}{P}$$

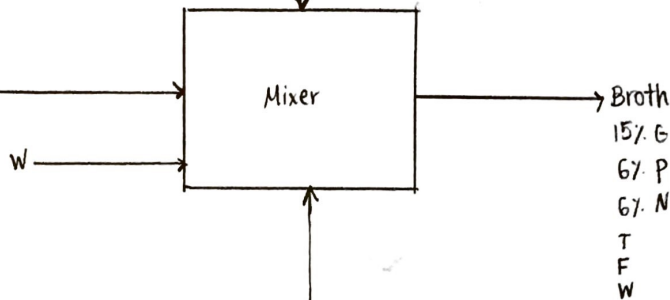
$$\dot{V} = \frac{(8000 \frac{\text{lbmol}}{\text{hr}}) \left(0.7302 \frac{\text{atm} \cdot \text{ft}^3}{\text{lbmol} \cdot ^\circ R} \right) (1440^\circ R)}{(100 \text{ atm})}$$

$$\dot{V} = 84119.04 \frac{\text{ft}^3}{\text{hr}}$$

"Fast Feed"
45% G
2% P
1% N
3% T
49% F

"Super-Gro"
0% G
35% P
15% N
8% T
42% F

"Formula N"
0% G
12% P
58% N
0% T
30% F



FF: Fast-Feed: \$20/kg
SG: Super-Grow: \$10/kg
FN: Formula-N: \$15/kg

Initial Composition for trace = 0.020 kg

FF = 0.333 kg
SG = 0.130 kg
FN = 0.0641 kg
G: 0.15 × 100 = 15
P: 0.059 × 100 = 5.9
N: 0.060 × 100 = 6.0
W: 0.473 × 100 = 47.3

Final Composition for trace = 0.030 kg

FF₂ = 0.333 kg
SG₂ = 0.25 kg
FN₂ = 0.0331 kg
W₂ = 0.384 kg
G: 0.15 × 100 = 15
P: 0.098 × 100 = 9.8
N: 0.060 × 100 = 6.0
W: 0.384 × 100 = 38.4

a) Material Balance for Glucose [1kg]

$$0.45FF = 0.15(1)$$

$$FF = 0.333 \text{ kg}$$

Material Balance for Phosphate [1kg]

$$0.02FF + 0.35SG + 0.12FN = 0.06(1)$$

$$0.02FF + 0.35SG + 0.12FN = 0.06$$

$$0.02(0.333) + 0.35SG + 0.12FN = 0.06$$

$$0.35SG + 0.12FN = 0.0533$$

Material Balance for Nitrate [1kg]

$$0.01FF + 0.15SG + 0.58FN = 0.06(1)$$

$$0.01FF + 0.15SG + 0.58FN = 0.06$$

$$0.01(0.333) + 0.15SG + 0.58FN = 0.06$$

$$0.15SG + 0.58FN = 0.0567$$

$$0.35SG + 0.12FN = 0.0533$$

$$0.15SG + 0.58FN = 0.0567$$

$$\begin{bmatrix} 0.35 & 0.12 \\ 0.15 & 0.58 \end{bmatrix} \begin{bmatrix} SG \\ FN \end{bmatrix} = \begin{bmatrix} 0.0533 \\ 0.0567 \end{bmatrix}$$

$$SG = 0.130 \text{ kg}$$

$$FN = 0.0641 \text{ kg}$$

$$FF + SG + FN + W = 1.0$$

$$0.333 \text{ kg} + 0.130 \text{ kg} + 0.0641 \text{ kg} + W = 1.0$$

$$W = 0.473 \text{ kg}$$

b) $FF(\text{kg}) \cdot \frac{\$}{\text{kg}} + SG(\text{kg}) \cdot \frac{\$}{\text{kg}} + FN(\text{kg}) \cdot \frac{\$}{\text{kg}} = \text{Total cost}$

$$0.333 \text{ kg} \left(\frac{\$20}{\text{kg}} \right) + 0.130 \text{ kg} \left(\frac{\$10}{\text{kg}} \right) + 0.0641 \text{ kg} \left(\frac{\$15}{\text{kg}} \right) = \text{Total cost}$$

$$\text{Total cost} = \$8.92/\text{kg}$$

$$\text{Savings} = \$15.00/\text{kg} - \$8.92/\text{kg} = \$6.08/\text{kg}$$

$$20 \text{ kg} \times \$6.08/\text{kg} = \$121.60 \text{ [savings for 20 kg]}$$

$$\frac{\$2000}{20 \text{ kg}}$$

$$\frac{\$121.60}{20 \text{ kg}}$$

$$= 16.45 \text{ batches}$$

$$\approx 17 \text{ batches}$$

[need to be made to pay for equipment]

c) Material Balance for Trace

$$0.03FF + 0.08SG + 0.0FN = \text{wt\% trace}$$

$$0.03(0.333) + 0.08(0.130) = 0.020 \text{ kg}$$

$$T = 0.020 \text{ kg}$$

Solving for new FN

$$0.15SG + 0.58FN = 0.0567$$

$$0.15(0.25) + 0.58FN = 0.0567$$

$$FN_2 = 0.0331 \text{ kg}$$

Mass Balance for Trace (T = 0.03 kg)

$$0.03FF + 0.08SG + 0.0FN = 0.03$$

$$0.03(0.333) + 0.08SG = 0.03$$

$$SG_2 = 0.25 \text{ kg}$$

$$FF + SG + FN + W = 1.0 \text{ [New SG and FN]}$$

$$0.333 \text{ kg} + 0.25 \text{ kg} + 0.0331 \text{ kg} + W = 1.0$$

$$W_2 = 0.384 \text{ kg}$$

Super-Grow was chosen because it has 8wt% trace (greater than 3 wt% requirement) There is no Formula N Trace nutrients.

$$0.333 \text{ kg} \left(\frac{\$20}{\text{kg}} \right) + 0.25 \text{ kg} \left(\frac{\$10}{\text{kg}} \right) + 0.0331 \text{ kg} \left(\frac{\$15}{\text{kg}} \right)$$

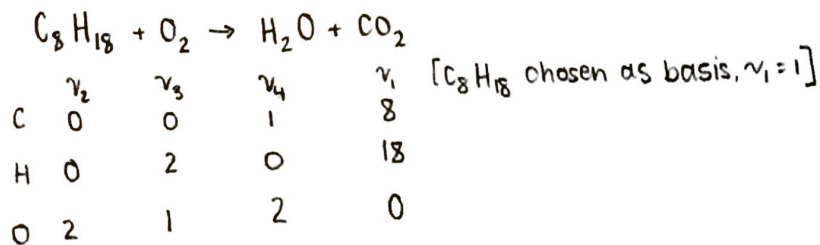
$$\text{New total cost: } \$9.66/\text{kg}$$

The new cost is greater but the specifications are met. However, \$9.66/kg is still less than \$15.00/kg. [The amount of P increased to ≈ 9 wt%.]

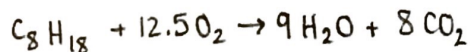
4.

CO₂ production per person in U.S.

$$\frac{0.25 [5.6 \times 10^9 \text{ metric tons}] \cdot \frac{2204.6 \text{ lb}}{1 \text{ metric ton}}}{300 \times 10^6 \text{ people}} = \boxed{10288.13 \text{ lb CO}_2 / \text{person [in one year]}}$$



$$\begin{bmatrix} \nu_2 \\ \nu_3 \\ \nu_4 \end{bmatrix} = \begin{bmatrix} -12.5 \\ 9 \\ 8 \end{bmatrix}$$



$$\text{C}_8\text{H}_{18} = 8(12.01) + 18(1.008) = 114.22 \text{ g/gmol} \quad \text{CO}_2 = 1(12.01) + 2(16.00) = 44.01 \text{ g/gmol}$$

$$\frac{8 \text{ gmol CO}_2 \times \frac{44.01 \text{ g CO}_2}{1 \text{ gmol CO}_2}}{1 \text{ gmol C}_8\text{H}_{18} \times \frac{114.22 \text{ g C}_8\text{H}_{18}}{1 \text{ gmol C}_8\text{H}_{18}}} = \frac{3.08 \text{ g CO}_2}{1 \text{ g C}_8\text{H}_{18}}$$

$$\frac{31.5 \text{ miles}}{\text{day}} \times \frac{365 \text{ days}}{1 \text{ year}} \times \frac{1 \text{ gal}}{27.5 \text{ miles}} \times \frac{3.78541 \text{ L}}{1 \text{ gal}} \times \frac{1000 \text{ mL}}{1 \text{ L}} \times \frac{0.7 \text{ g C}_8\text{H}_{18}}{1 \text{ mL}} \times \frac{3.08 \text{ g CO}_2}{1 \text{ g C}_8\text{H}_{18}} \times \frac{1 \text{ lb}}{453.59 \text{ g}} = \boxed{7522 \text{ lb CO}_2 / \text{year}}$$

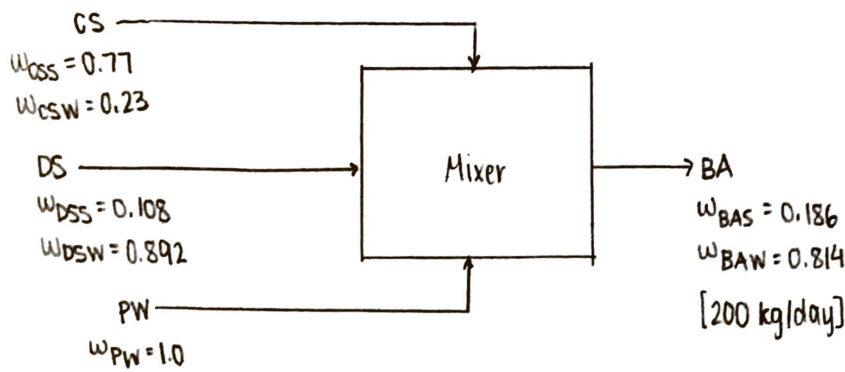
(7522 lb CO₂/year)

The value is reasonable since it is less than 10288.13 lb CO₂/year and only accounts for driving, CO₂ is released other ways as well.

31.5 miles/day [aaaFoundation.org, American Driving Survey, 2015-2016] ***

↳ This value was looked up.

20/20



CS: Concentrated Solution
 CSS: Concentrated Solution Sulfuric
 CSW: concentrated Solution Water
 DS: Diluted Solution
 DSS: Diluted solution Sulfuric
 DSW: Diluted Solution Water
 PW: Pure Water
 BA: Battery Acid
 BAS: Battery Acid Sulfuric
 BAW: Battery Acid Water

DOF Analysis

stream var: 7 specified flow: 1
 system var: 0 specified stream comp: 3
 7 specified system perf: 0
 material balance eqns: 2

$$DOF = 7 - 6 = 1 \text{ [underspecified]}$$

Material Balance for Sulfuric Acid

$$0.77CS + 0.108DS + 0.0PW = 0.186 [200 \text{ kg/day}]$$

$$0.77CS + 0.108DS = 37.2 \text{ kg/day}$$

Material Balance for Water

$$0.23CS + 0.892DS + 1.0PW = 0.814 [200 \text{ kg/day}]$$

$$0.23CS + 0.892DS + 1.0PW = 162.8 \text{ kg/day}$$

Cost Balance

$$0.10CS + 0.02DS = \text{cost} [\$/\text{day}]$$

Solve

$$\frac{\$0.10}{0.77} = \$0.130/\text{kg} \rightarrow \text{cheaper} \quad [\text{CS}]$$

$$\frac{\$0.02}{0.108} = \$0.185/\text{kg} \rightarrow \text{DS} = 0 \text{ kg/day} \quad [\text{DS}] \quad \text{[by completely eliminating DS, the cheaper option is to only use the concentrated CS]}$$

$$0.77CS + 0.108DS = 37.2$$

$$0.77CS + 0.108(0) = 37.2$$

$$CS = 48.31 \text{ kg/day}$$

$$0.23CS + 0.892DS + 1.0PW = 162.8$$

$$0.23(48.31) + 0.892(0) + 1.0PW = 162.8$$

$$PW = 151.69 \text{ kg/day}$$

Flow Rates

CS: 48.31 kg/day
 DS: 0 kg/day
 PW: 151.69 kg/day

$$CS + DS + PW = BA$$

$$48.31 + 0 + 151.69 = 200$$

$$200 \checkmark 200 \text{ kg/day}$$

20
 20